

Surfaces that Minimize Area

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Abstract

In this report, I aim to present the reader with several equivalent definitions of minimal surfaces. To do this, I will use the calculus of variations to seek solutions for the Plateau problem with the solutions to the area minimization problem for given boundaries are minimal surfaces. I will also explain why we constructed the variations in the way we did to solve the Plateau problem.

Furthermore, I will use differential geometry to understand the general nature of minimal surfaces, including those that are not area minimizers and therefore do not solve the Plateau problem. Throughout this process, I will proceed by analogies on geodesics, which are the 1-dimensional equivalents of minimal surfaces. Finally, I will present some examples of minimal surfaces and open questions.

1 Introduction

Minimal surfaces originally comes from the search for an area-minimizing surface for a given boundary curve. This is the "Plateau's Problem". To understand minimal surfaces, we will first use the calculus of variations to reach an equivalent definition. This definition states that the Mean curvature is zero at every point.

But before that, using only this property, we will learn about their Gaussian Curvature and how the points on that surface behave and it's possibility of being compact.

From this, it will become clear that minimal surfaces cannot be compact because they cannot contain an elliptic point. However, the surface we are trying to understand in the calculus of variations is definitely bounded. As stated in Pressley, shortest path geodesics on closed subsets can be extended indefinitely, likewise we'll be talking about extended versions of minimal surfaces(without boundary).

Considering that geodesics are the 1-dimensional analogue of minimal surfaces, we should state this clearly: given boundary points on a surface in , the shortest path is a geodesic, but it does not necessarily exist (for example, a plane without the origin). Likewise, Existence of area-minimizing minimal surfaces was a problem that solved by Douglas-Rado independently.

We need to prevent potential confusion here as well. The calculus of variations does not tell us if a surface is an area minimizer; it only tells us that it is a critical point of the area functional. In other words, it gives us local information. This makes more sense when we consider that geodesics with given boundary points do not always provide the shortest path.

In the case of minimal surfaces, we will check if the area function does change when we take a variation of the surface, bounded by given curve.

2 Minimal surfaces

Recalling the shortest path problem for given boundary points on a surface in $\mathbb{R}^3$, we observe that while every shortest path is a geodesic, the converse is not generally true. Any helix on a cylinder serves as an example of this. Similarly, not every minimal surface is an area minimizer. In fact, much like geodesics, they are only the local critical points of the area function.

On existence, Douglas and Rado proved that for every simple closed curve in $\mathbb{R}^3$, there exists a disk-type minimal surface. The non-uniqueness of geodesics can be observed on a sphere by choosing two points that divide a great circle into two equal lengths. To show, minimal surfaces are also not unique, we will later present the example of the Catenoid and the Goldschmidt solution. However, it remains an open question whether there exists a curve that can span uncountably many disk-type minimal surfaces!

3 Local Geometry of Minimal Surfaces

Before exploring minimal surfaces, it is important to understand the local behavior of surfaces in $\mathbb{R}^3$.

Definition A surface $S \subset \mathbb{R}^3$ is *minimal* if and only if its mean curvature vanishes identically.

Since the **Mean Curvature** is zero everywhere, we can understand the shape of the surface using **Gaussian curvature**. (For simplicity, we will momentarily neglect the case where $k_1 = k_2 = 0$, i.e., planar points, though it is worth noting that minimal surfaces like Enneper's surface do indeed contain planar points but in not isolated form is only the plane).

From the equation $H = \frac{k_1+k_2}{2} = 0$, it follows that $k_1 = -k_2$. Consequently, $K = k_1k_2 \leq 0$. This implies that every point is either **hyperbolic** or **planar**; there are no elliptic or parabolic points.

For hyperbolic points, observe that at a point p , we can choose two curves passing through p such that they curve away towards opposite sides of the tangent plane at p . One should immediately visualize surfaces that consist **saddle points**!

To be more precise, we need the proposition on the **local position** of the surface relative to its tangent plane.

Proposition 3.1 (Local Position). *Let $p \in S$ be a point on a regular surface.*

1. *If p is an **elliptic point** ($K > 0$), there exists a neighborhood V of p in S such that all points in V belong to the same side of the tangent plane $T_p(S)$.*
2. *If p is a **hyperbolic point** ($K < 0$), then in every neighborhood of p , there exist points of S on both sides of $T_p(S)$.*

The proof of this proposition can be found in Do Carmo. I would like to draw attention to the case of *elliptic points*. This is because an elliptic point indicates that the surface have local bump extremas in R^3 . In fact, for a surface to be Compact it must necessarily contain elliptic points!

Proposition 3.2 (Compact surfaces). *A regular compact surface(without boundary) $S \subset \mathbb{R}^3$ has at least one elliptic point.*

Proof. Since S is compact, S is bounded. Therefore, there are spheres of $\mathbb{R}^3$, centered in a fixed point $O \in \mathbb{R}^3$, such that S is contained in the interior of the region bounded by any of them. Consider the set of all such spheres. Let r be the infimum of their radii and let $\Sigma \subset \mathbb{R}^3$ be a sphere of radius r centered in O . It is clear that Σ and S have at least one common point, say p . The tangent plane to Σ at p has only the common point p with S , in a neighborhood of p . Therefore, Σ and S are tangent at p . By observing the normal sections at p , it is easy to conclude that any normal curvature of S at p is greater than or equal to the corresponding curvature of Σ at p . Therefore, $K_S(p) \geq K_\Sigma(p) > 0$, and p is an elliptic point, as we wished. **Q.E.D.**

Remark 3.3. Now, there is no minimal surface which is compact, as they contain no elliptic points and are therefore unbounded. Consequently, they cannot be compact. At this point, we will adopt a slightly confusing approach because, conventionally, we stated that the historical method for obtaining minimal surfaces involves finding surfaces that span given simple closed curves as boundaries. I would like to refer to these as “minimal surfaces with boundary.” In reality, we are not doing anything fundamentally different; we can think of it as starting by obtaining smaller minimal surfaces (with boundary) to eventually understand complete minimal surfaces. This is analogous to finding a segment of a geodesic when using the shortest path found while searching for geodesics. Furthermore, after you find some shortest path(geodesic) these can be extended beyond the line segment.

Now, let us explain why the proof we constructed is not valid for the case minimal surfaces with boundary points. The sphere Σ touches our surface at an *interior point*! If the contact point were on the boundary, we could not define a tangent space there. Consequently, we could not discuss normal curvature, the surface normal we need.

4 Variational Formulation

So far, we have discussed the Plateau problem and said that its solutions take the form of minimal surfaces. We said that these are not “minimal surfaces” in the most general sense (i.e., those without boundary).

We will now introduce a method to test whether a surface is a candidate for being an area minimizer: the so-called **First Variational Formula**. Before that, let us first recall the analogous case for geodesics. Geodesics are critical points of length function among all variations.

Let $\alpha : [a, b] \rightarrow S$ be a curve on the surface. We choose a variation vector field $V(t)$ along α such that $V(a) = V(b) = 0$ (We slightly vibrated the curve except for the boundary

points). The family of curves α_s is defined as:

$$\alpha(t, s) = \alpha(t) + sV(t).$$

Here, $s \in (-\epsilon, \epsilon)$ is the variation parameter. first variation of the length $L(\alpha)$ is:

$$L(\alpha) = \int_a^b |\alpha'(t)| dt \implies L'(0) = - \int_a^b \left\langle \frac{D\alpha'}{dt}, V(t) \right\rangle dt$$

Now we can show the same for dimension 2, i.e Minimal Surfaces.

4.1 The Area Functional

Let $\mathbf{x} : U \subset \mathbb{R}^2 \rightarrow S \subset \mathbb{R}^3$ be a parametrization of a surface, let $D \subset U$ be a bounded domain, and let $h : \tilde{D} \rightarrow \mathbb{R}$ be a differentiable function vanishing on the boundary. (variations have the same boundary as before, but even if it did not still would be true. So the results hold the same on the surface variation φ which is normal to the surface along the boundary curve.) The *normal variation* of $\mathbf{x}(\tilde{D})$, determined by h , is the map $\varphi : \tilde{D} \times (-\epsilon, \epsilon) \rightarrow \mathbb{R}^3$:

$$\varphi(u, v, t) = \mathbf{x}(u, v) + th(u, v)\mathbf{N}(u, v).$$

For fixed t , the map $\mathbf{x}^t(u, v) = \varphi(u, v, t)$ is a parametrized regular surface with

$$\begin{aligned} \frac{\partial \mathbf{x}^t}{\partial u} &= \mathbf{x}_u + thN_u + th_uN, \\ \frac{\partial \mathbf{x}^t}{\partial v} &= \mathbf{x}_v + thN_v + th_vN. \end{aligned}$$

Using the orthogonality condition $\langle \mathbf{x}_u, \mathbf{N} \rangle = \langle \mathbf{x}_v, \mathbf{N} \rangle = 0$, the coefficients expand as follows:

$$\begin{aligned} E^t &= E + th(\langle \mathbf{x}_u, N_u \rangle + \langle \mathbf{x}_u, N_u \rangle) + t^2h^2\langle N_u, N_u \rangle + t^2h_uh_u, \\ F^t &= F + th(\langle \mathbf{x}_u, N_v \rangle + \langle \mathbf{x}_v, N_u \rangle) + t^2h^2\langle N_u, N_v \rangle + t^2h_uh_v, \\ G^t &= G + th(\langle \mathbf{x}_v, N_v \rangle + \langle \mathbf{x}_v, N_v \rangle) + t^2h^2\langle N_v, N_v \rangle + t^2h_vh_v. \end{aligned}$$

Recall that the area functional is $A(t) = \int_{\tilde{D}} \sqrt{E^tG^t - (F^t)^2} dudv$.

Coefficients of the second fundamental form of original surface: $-e = \langle \mathbf{x}_u, \mathbf{N}_u \rangle$, $-f = \langle \mathbf{x}_u, \mathbf{N}_v \rangle = \langle \mathbf{x}_v, \mathbf{N}_u \rangle$, and $-g = \langle \mathbf{x}_v, \mathbf{N}_v \rangle$. Now to calculate the area of variation:

$$\begin{aligned} E^tG^t - (F^t)^2 &= (E - 2the)(G - 2thg) - (F - 2thg)^2 + R.(t^2) \\ &= (EG - F^2) - 2th(Eg + Gf - 2Fg) + R.(t^2). \end{aligned}$$

Recalling the definition of Mean Curvature $H = \frac{1}{2} \frac{Eg - 2fF + Ge}{EG - F^2}$ we can factor out $EG - F^2$:

$$\begin{aligned} E^tG^t - (F^t)^2 &= (EG - F^2) \left[1 - \frac{2th(Eg + Ge - 2Fg)}{EG - F^2} \right] + R.(t^2) \\ &= (EG - F^2)(1 - 4thH) + R.(t^2). \end{aligned}$$

Since the limit of the remainder term R/t goes to 0 as $t \rightarrow 0$, it follows that if ϵ is sufficiently small, the determinant remains positive:

$$E^t G^t - (F^t)^2 > 0.$$

Thus, for small ϵ , $\mathbf{x}^t$ is a regular parametrized surface.

Furthermore, the area $A(t)$ of the perturbed surface $\mathbf{x}^t(\bar{D})$ is given by:

$$\begin{aligned} A(t) &= \int_{\bar{D}} \sqrt{E^t G^t - (F^t)^2} \, du \, dv \\ &= \int_{\bar{D}} \sqrt{1 - 4thH + \bar{R}\sqrt{EG - F^2}} \, du \, dv, \end{aligned}$$

where $\bar{R} = R/(EG - F^2)$.

It follows that if ϵ is small, $A(t)$ is a differentiable function of t . By differentiating under the integral sign at $t = 0$, we obtain the First Variation of Area:

$$A'(0) = - \int_{\bar{D}} 2hH\sqrt{EG - F^2} \, du \, dv.$$

We are now in a position to characterize area-minimizing surfaces.

Theorem 4.1. *Let $\mathbf{x} : D \rightarrow \mathbb{R}^3$ be a regular parametrized surface. A necessary condition for $\mathbf{x}$ to be a critical point of the area function (i.e., $A'(0) = 0$ for all variations) is that the mean curvature H vanishes each point, being minimal surface.*

Remark: Surfaces with $H \equiv 0$ are called *minimal surfaces*. Historically, this name derives from the fact that they are the critical points for the area minimization problem.

It is important to notice that this critical point is not always a minimum (the smallest possible area). This makes the name “minimal surface” seem a bit confusing. However, we still use this name because it is a long-standing tradition started by Lagrange, who first defined minimal surfaces in 1760.

Proposition A critical point for the area functional is not necessarily a minimum. A way to justify the word *minimal* is by asking that the surface S is **stable**. Stability is studied with respect to the **second variation**: ($A''(0) = 0$)

A "stable" minimal surface corresponds to a local minimum of the area function. It means that if you perturb the surface slightly, the area will increase. A critical point is not necessarily a minimum.

5 Mean Curvature Vector

It is useful to define the **mean curvature vector** as $\mathbf{H} = H\mathbf{N}$ for any regular surface.

If we choose our deformation to be exactly $h = H$, the formula gives:

$$A'(0) = -2 \int_{\bar{D}} \langle \mathbf{H}, \mathbf{H} \rangle \sqrt{EG - F^2} \, du \, dv < 0.$$

This result shows that **if we push the surface in the direction of $\mathbf{H}$, the area starts to decrease.**

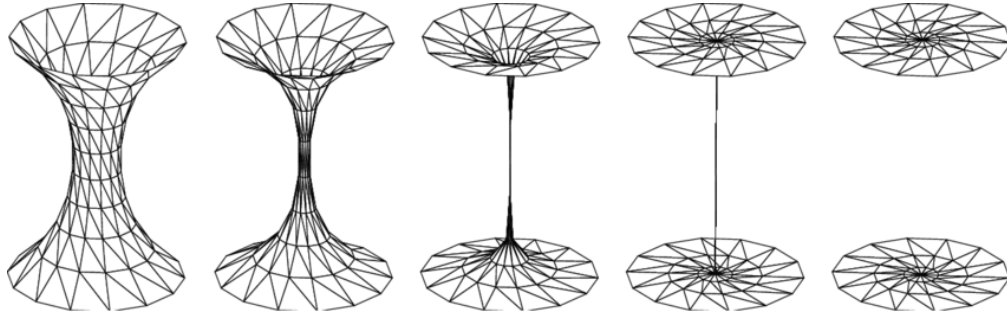
You can think of $\mathbf{H}$ as a **pressure difference** across the surface. Just as pressure forces a soap film to bend and minimize its area, the vector $\mathbf{H}$ points in the direction the surface "wants" to move to reduce its tension.

6 Uniqueness

Uniqueness holds under strict constraints.

Theorem 6.1 (Radó's Uniqueness Condition). *If the Jordan curve Γ admits a one-to-one projection onto a convex plane curve, then it bounds a unique minimal surface of disc-type.*

A great example of non-uniqueness is Catenoid and Goldschmidt solution(discontinuous) that bounded by 2 circle.



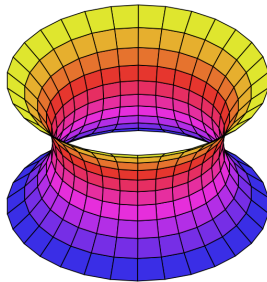
7 The Catenoid

It is the only minimal surface of revolution (excluding the trivial plane).

$$\mathbf{x}(u, v) = (a \cosh v \cos u, a \cosh v \sin u, av)$$

where $u \in [0, 2\pi)$ $v \in \mathbb{R}$.

Boundary Bounded two circular wire rings. If the rings are separated beyond a critical distance, the film becomes unstable and collapses into two disjoint planar discs (the Goldschmidt solution).



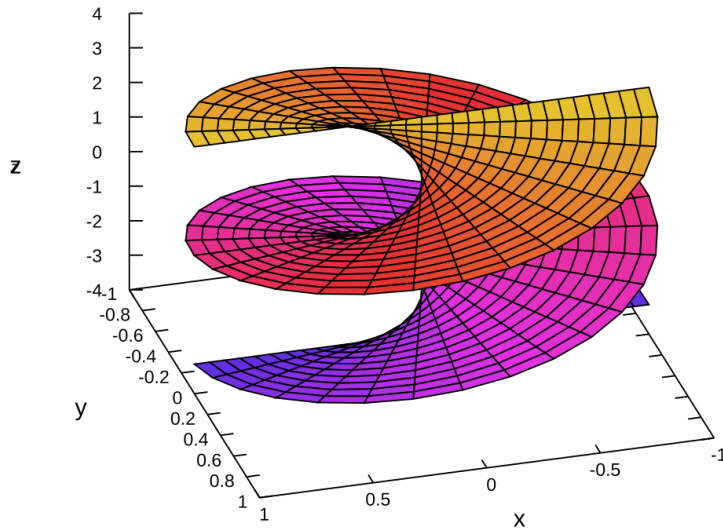
7.1 The Helicoid

The helicoid is a ruled surface, generated by a straight line moving along a curve. It is locally isometric to the catenoid.

$$\mathbf{x}(u, v) = (u \cos v, u \sin v, av)$$

where $u \in [0, 2\pi)$ $v \in \mathbb{R}$.

Boundary: Circular helix.



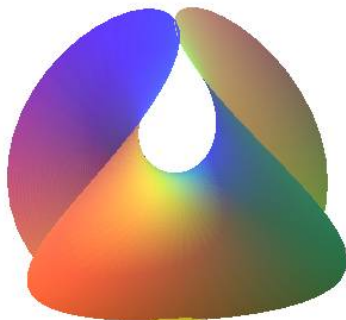
7.2 Enneper's Surface

Enneper's is defined entirely by polynomials.

$$\mathbf{x}(u, v) = \left(u - \frac{u^3}{3} + uv^2, \quad v - \frac{v^3}{3} + vu^2, \quad u^2 - v^2 \right)$$

where $u, v \in \mathbb{R}$.

Boundary: Enneper's surface **cannot** be realized as a standard, non-intersecting soap film in physical space.



8 The Finiteness Problem

We know that a specific boundary curve Γ can bound more than one minimal surface. However, it is not known if the number of such surfaces is always finite.

Open Problem 8.1. Does every real analytic Jordan curve in $\mathbb{R}^3$ bound only a finite number of minimal surfaces of disk-type?

References

- [1] J. Douglas, *Solution of the problem of Plateau*, Trans. Amer. Math. Soc. **33** (1931), 263–321.
- [2] J. Harrison and H. Pugh, *Plateau’s Problem: What’s Next*, arXiv preprint arXiv:1509.03797v2 [math.AP], 2016.
- [3] R. Osserman, *A Survey of Minimal Surfaces*, Van Nostrand Reinhold, New York, 1969.
- [4] M. P. do Carmo, *Differential Geometry of Curves and Surfaces*, Prentice-Hall, 1976.
- [5] A. Pressley, *Elementary Differential Geometry*, Springer, 2010.
- [6] U. Pinkall and K. Polthier, *Computing Discrete Minimal Surfaces and Their Conjugates*, Experimental Mathematics **2** (1993), no. 1, 15–36.